NLPrompt: Noise-Label Prompt Learning for Vision-Language Models

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Abstract

Prompt learning has emerged as a parameter-efficient strategy for adapting vision-language models such as CLIP to downstream tasks, yet its effectiveness degrades substantially when training labels are noisy. We observe that the standard cross-entropy (CE) loss used in prompt tuning is highly susceptible to overfitting mislabeled samples, a problem amplified in few-shot regimes where every annotation carries outsized influence. In this work, we propose NLPrompt, a unified framework for noise-robust prompt learning that integrates two complementary components. First, PromptMAE replaces CE with the mean absolute error (MAE) loss during prompt optimization, which we show both empirically and theoretically attenuates the gradient contribution of noisy samples and increases the signal-to-noise ratio in learned features. Second, PromptOT leverages optimal transport between image features and text-feature prototypes derived from the learned prompts to globally partition the training set into clean and noisy subsets. NLPrompt then harmonizes CE training on the identified clean subset with MAE training on the noisy subset, exploiting the strengths of both losses. Extensive experiments on seven benchmarks under both symmetric and asymmetric label noise demonstrate that NLPrompt consistently outperforms existing methods by large margins. For instance, on Flowers102 with 75% symmetric noise NLPrompt achieves 76.80% accuracy compared to 70.37% for GCE and 66.93% for JoAPR. On the real-world noisy dataset Food101N, NLPrompt reaches 76.46%, surpassing all baselines. Furthermore, our framework generalizes seamlessly to other prompt tuning methods including VPT, MaPLe, and PromptSRC.

1. Introduction

Vision-language models (VLMs) pre-trained on web-scale image-text pairs, such as CLIP [35], have demonstrated remarkable zero-shot recognition capabilities. Prompt learning [45, 46] further unlocks efficient adaptation of these models to downstream tasks by optimizing a small set of continuous context tokens while keeping the pre-trained backbone frozen. This paradigm has achieved impressive few-shot performance across diverse visual recognition benchmarks.

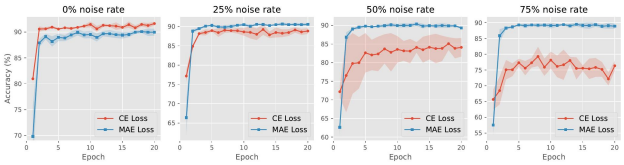

Figure 1. Test accuracy on Caltech101 across training epochs under different symmetric noise rates. CE loss (red) degrades sharply as noise increases, while MAE loss (blue) maintains robust performance. At 75% noise, MAE surpasses CE by a wide margin, motivating our PromptMAE component.

However, real-world datasets frequently contain noisy labels arising from crowd-sourced annotation, web crawling, or automated labeling pipelines [7, 43]. While the memorization effects of deep networks under noisy labels have been extensively studied in conventional training [16, 25], the impact on prompt learning remains relatively under-explored. Recent work [40] has observed that prompt tuning exhibits some inherent robustness to label noise due to the regularization provided by fixed class-name tokens. Nevertheless, this robustness is limited: as the noise ratio increases, prompt tuning with the standard cross-entropy (CE) loss still overfits to corrupted labels, leading to severe performance degradation (see Figure 1).

A natural question arises: *can we design a prompt learning framework that is inherently robust to label noise?* To answer this, we first revisit the choice of loss function. Prior work in conventional deep learning has established that the mean absolute error (MAE) is theoretically robust to label noise [14], yet it has seen limited adoption in prompt learning. We demonstrate that replacing CE with MAE in prompt tuning—a method we term *PromptMAE*—significantly improves robustness under noisy labels. Through theoretical analysis grounded in feature learning theory [19, 39], we prove that MAE reduces the influence of noisy samples on the learned prompt features by improving the signal-to-noise ratio during optimization.

While PromptMAE provides strong noise robustness, CE loss remains advantageous on clean data due to its faster convergence and tighter decision boundaries. To leverage

both losses, we introduce *PromptOT*, an optimal-transport-based data purification module. PromptOT constructs a cost matrix from the negative log-similarity between image features and text-feature prototypes generated by the current prompts, then solves an entropic optimal transport problem [10] to obtain globally consistent pseudo-labels. By comparing these pseudo-labels with the given noisy annotations, PromptOT partitions the dataset into a clean subset and a noisy subset. A key advantage of PromptOT over existing sample selection methods is its use of text features as class prototypes, which provides semantically grounded anchors that do not drift with noisy training signals.

Our complete framework, *NLPrompt*, harmonizes CE training on the clean subset with MAE training on the noisy subset, combining the discriminative power of CE with the noise tolerance of MAE. Figure 2 illustrates the overall pipeline. The contributions of this paper are summarized as follows:

- We identify and validate PromptMAE: the use of MAE loss as a simple yet highly effective strategy for noise-robust prompt learning in vision-language models.
- We provide a theoretical analysis via feature learning theory showing that MAE reduces the influence of noisy samples on prompt optimization, leading to improved generalization over CE.
- We propose PromptOT, an optimal-transport-based data purification method that leverages text-feature prototypes for globally consistent pseudo-labeling and clean/noisy data separation.
- We introduce NLPrompt, which harmonizes CE and MAE on separated clean and noisy subsets, achieving state-of-the-art robustness across seven benchmarks with both synthetic and real-world noise. NLPrompt also generalizes to other prompt tuning methods such as VPT [20], MaPLe [21], and PromptSRC [22].

2. Related Work

2.1. Prompt Learning for Vision-Language Models

Large-scale vision-language models such as CLIP [35] learn aligned image-text representations through contrastive pre-training on hundreds of millions of image-text pairs. To adapt these models to downstream tasks, CoOp [45] proposes to replace hand-crafted text prompts with learnable continuous context vectors, achieving significant gains over zero-shot baselines with only a few labeled examples. CoCoOp [46] extends this idea by conditioning the prompt on each input image to improve generalization to unseen classes. MaPLe [21] introduces multi-modal prompt learning that jointly optimizes prompts in both the vision and language branches. PromptSRC [22] adds self-regularization constraints to prevent forgetting the pre-trained knowledge during prompt tuning. ProGrad [47] aligns prompt gradients with zero-shot predictions to preserve generalization. Visual Prompt Tuning (VPT) [20] and test-time prompt tuning (TPT) [36] explore prompting in the visual encoder and at test time, respectively. PLOT [6] uses optimal transport to align multiple text prompts with visual features. While these methods have advanced prompt learning considerably, they generally assume clean training labels and do not address the challenges posed by label noise.

2.2. Learning with Noisy Labels

Learning with noisy labels (LNL) has been extensively studied in conventional deep learning. One prominent line of work focuses on noise-robust loss functions. Ghosh et al. [14] theoretically establish that MAE is inherently robust to label noise under certain conditions, while CE is not. Subsequent works propose generalized cross-entropy (GCE) [13], symmetric cross-entropy [38], normalized losses [27], and curriculum loss [26] to balance robustness and learnability.

Another line of work addresses noisy labels through sample selection and data purification. Co-teaching [16] trains two networks that teach each other by selecting small-loss samples. DivideMix [25] models the per-sample loss distribution with a Gaussian mixture model to separate clean and noisy samples, then applies semi-supervised learning. CORES^2 [8] proposes a confidence-regularized sample sieve for instance-dependent noise. Transition matrix methods [33, 42, 44] estimate the noise transition probabilities for loss correction.

Optimal transport has also been applied to LNL. OT-Filter [12] uses OT to filter noisy samples based on feature-space distances. OT-Cleaner [41] formulates label correction as an OT problem. CSOT [5] proposes a curriculum and structure-aware OT formulation for robust denoising. In the semi-supervised and pseudo-labeling literature, Sinkhorn-based allocation has been used for balanced label assignment [2, 4, 29, 37].

2.3. Prompt Learning under Label Noise

The intersection of prompt learning and noisy labels has received limited attention. Wu et al. [40] analyze why prompt tuning is relatively robust to label noise compared to full fine-tuning, attributing this to the regularization effect of fixed class-name tokens and the strong prior from pre-trained representations. JoAPR [15] proposes a joint framework that refines prompts and purifies labels simultaneously using agreement between text and visual predictions. However, existing approaches either do not fully exploit robust loss functions or lack principled mechanisms for separating clean and noisy data in the prompt learning setting. NLPrompt addresses both gaps by integrating MAE-based robust training with OT-based data purification, providing a theoretically grounded and empirically effective

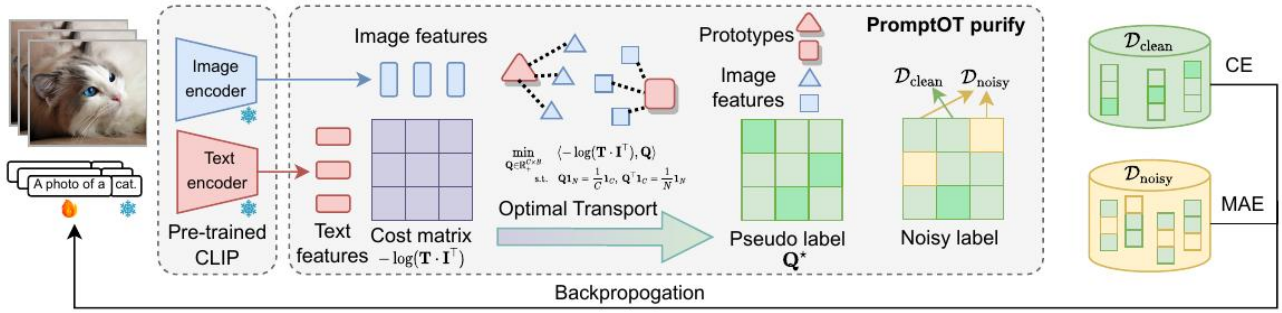

Figure 2. Overview of NLPrompt. Given a pre-trained CLIP model with learnable prompt tokens, we extract image features and text features to construct a cost matrix based on negative log-similarity. An optimal transport plan is computed via the Sinkhorn algorithm to produce pseudo-labels. By comparing these with the given noisy labels, PromptOT separates the data into a clean subset $\mathcal{D}_{\text{clean}}$ trained with CE loss and a noisy subset $\mathcal{D}_{\text{noisy}}$ trained with MAE loss. Gradients from both branches update the shared prompt tokens.

framework.

3. Preliminary

3.1. Prompt Learning with CLIP

CLIP [35] consists of an image encoder f_I and a text encoder f_T trained to align visual and textual representations through contrastive learning. For a classification task with C classes, a text prompt "a photo of a [CLASS]" is processed by the text encoder to produce class-wise text features $\{\boldsymbol{t}_c\}_{c=1}^{C}$. Given an image $\boldsymbol{x}$ with image feature $\boldsymbol{z} = f_I(\boldsymbol{x})$, the prediction probability for class c is computed as:

$$p(y = c \mid \boldsymbol{x}) = \frac{\exp(\text{sim}(\boldsymbol{z}, \boldsymbol{t}_c)/\tau)}{\sum_{j=1}^{C} \exp(\text{sim}(\boldsymbol{z}, \boldsymbol{t}_j)/\tau)}, \tag{1}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity and τ is a learned temperature parameter.

CoOp [45] replaces the hand-crafted prompt with learnable context vectors $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_M\}$, forming the prompt as $[\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_M, \text{CLASS}_c]$ for each class c. Only the context vectors are optimized while the entire CLIP backbone remains frozen, yielding a parameter-efficient adaptation strategy.

3.2. Label Noise Model

Let $\mathcal{D} = \{(\boldsymbol{x}_i, \tilde{y}_i)\}_{i=1}^{N}$ denote the training set where $\tilde{y}_i$ is the observed (potentially noisy) label and y_i^* is the unknown clean label. Under symmetric noise with rate η, each label is independently flipped to a uniformly random class with probability η. Under asymmetric noise, labels are flipped only to semantically similar classes. The standard prompt learning objective minimizes the CE loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N} \log p(\tilde{y}_i \mid \boldsymbol{x}_i). \tag{2}$$

When η is large, minimizing Eq. (2) causes the prompt to overfit to corrupted labels, degrading generalization.

3.3. Robustness of MAE Loss

The mean absolute error (MAE) loss for classification is defined as:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{N}\sum_{i=1}^{N} \left(1 - p(\tilde{y}_i \mid \boldsymbol{x}_i)\right). \tag{3}$$

Ghosh et al. [14] prove that MAE satisfies a noise-tolerance condition: for any loss ℓ satisfying $\sum_{c=1}^{C} \ell(f(\boldsymbol{x}), c) = R$ for a constant R and all $\boldsymbol{x}$, the risk minimizer under noisy labels coincides with that under clean labels. MAE satisfies this condition since $\sum_{c=1}^{C}(1 - p(c \mid \boldsymbol{x})) = C-1$ for all $\boldsymbol{x}$. In practice, we implement MAE through the generalized cross-entropy (GCE) [13] with parameter $q = 1$, which reduces to MAE:

$$\mathcal{L}_{\text{GCE}}(q) = \frac{1 - p(\tilde{y}_i \mid \boldsymbol{x}_i)^q}{q}. \tag{4}$$

When $q \to 0$, GCE recovers CE; when $q = 1$, GCE becomes MAE.

4. Theoretical Analysis for the Robustness of PromptMAE

In this section, we provide a theoretical analysis of why replacing CE with MAE in prompt learning improves robustness to label noise, building on feature learning theory [1, 3, 19, 39].

4.1. Data Model

Following the feature learning framework [18, 19], we consider a simplified data generation model. Each sample $(\boldsymbol{x}, y)$ consists of a signal component aligned with the true

class and a noise component. Specifically, the input $\boldsymbol{x}$ can be decomposed as:

$$\boldsymbol{x} = \underbrace{\alpha \cdot \boldsymbol{\mu}_y}_{\text{signal}} + \underbrace{\boldsymbol{\xi}}_{\text{noise}}, \tag{5}$$

where $\boldsymbol{\mu}_y$ is the class-specific feature pattern, $\alpha > 0$ controls the signal strength, and $\boldsymbol{\xi}$ is a noise vector. The signal-to-noise ratio (SNR) is defined as $\text{SNR} = \alpha / \|\boldsymbol{\xi}\|$.

Under label noise with rate η, the observed label $\tilde{y}$ differs from the true label y^* with probability η. This effectively corrupts the association between $\boldsymbol{\mu}_y$ and the observed label.

4.2. Gradient Analysis

We analyze the gradient contributions from clean and noisy samples under CE and MAE losses. For a prompt parameter $\boldsymbol{v}$ and a sample $(\boldsymbol{x}_i, \tilde{y}_i)$, the gradient of the CE loss with respect to $\boldsymbol{v}$ is proportional to:

$$\nabla_{\boldsymbol{v}} \mathcal{L}_{\text{CE}} \propto (p(\tilde{y}_i \mid \boldsymbol{x}_i) - 1) \cdot \frac{\partial \boldsymbol{t}_{\tilde{y}_i}}{\partial \boldsymbol{v}} \cdot \boldsymbol{z}_i, \tag{6}$$

where the factor $(p(\tilde{y}_i \mid \boldsymbol{x}_i) - 1)$ does not saturate and can have large magnitude for mislabeled samples with confident wrong predictions.

In contrast, the gradient of the MAE loss satisfies:

$$\nabla_{\boldsymbol{v}} \mathcal{L}_{\text{MAE}} \propto -p(\tilde{y}_i \mid \boldsymbol{x}_i)(1 - p(\tilde{y}_i \mid \boldsymbol{x}_i)) \cdot \frac{\partial \boldsymbol{t}_{\tilde{y}_i}}{\partial \boldsymbol{v}} \cdot \boldsymbol{z}_i. \tag{7}$$

The key difference is the multiplicative factor $p(\tilde{y}_i \mid \boldsymbol{x}_i)(1 - p(\tilde{y}_i \mid \boldsymbol{x}_i))$, which is bounded by $1/4$ and approaches zero when the model is confident (either correctly or incorrectly). This implicit gradient clipping [28] suppresses the influence of noisy samples on the prompt update.

4.3. Signal-to-Noise Ratio Improvement

Building on the feature learning analysis of [18, 19], we can characterize the learning dynamics of the prompt parameters. Consider the decomposition of the prompt gradient into a signal term (from correctly labeled samples) and a noise term (from mislabeled samples).

Theorem 4.1 (Informal)**.** *Under the data model described above and standard assumptions on the network architecture, training with MAE loss achieves a higher effective signal-to-noise ratio in the prompt feature space compared to CE loss. Specifically, the ratio of the signal gradient magnitude to the noise gradient magnitude satisfies:*

$$\frac{\|\nabla^{MAE}_{signal}\|}{\|\nabla^{MAE}_{noise}\|} \geq \frac{\|\nabla^{CE}_{signal}\|}{\|\nabla^{CE}_{noise}\|}, \tag{8}$$

where the inequality becomes strict when there exist confidently mislabeled samples.

The intuition is that MAE's bounded gradient property uniformly suppresses all gradient magnitudes, but the suppression is proportionally stronger for noisy samples (which tend to have confident but incorrect predictions) than for clean samples. This results in the prompt learning process being guided more strongly by the true signal in the data, leading to better generalization. This analysis is supported by the empirical observations in Figure 1, where MAE consistently outperforms CE under high noise rates despite slightly slower convergence on clean data.

5. Methodology

5.1. PromptMAE: Robust Prompt Learning with MAE Loss

Motivated by the theoretical analysis in the preceding section and the empirical observations in Figure 1, we propose PromptMAE as a drop-in replacement for CE in prompt learning. PromptMAE optimizes the learnable context vectors $\{\boldsymbol{v}_m\}_{m=1}^{M}$ by minimizing the MAE loss:

$$\min_{\{\boldsymbol{v}_m\}} \frac{1}{N} \sum_{i=1}^{N} \left(1 - p(\tilde{y}_i \mid \boldsymbol{x}_i; \{\boldsymbol{v}_m\})\right), \tag{9}$$

where $p(\tilde{y}_i \mid \boldsymbol{x}_i; \{\boldsymbol{v}_m\})$ is computed according to Eq. (1) using the learned prompts. PromptMAE retains the parameter efficiency of CoOp—only the context vectors are updated—while inheriting the noise-tolerance guarantees of MAE. As shown in Figure 1, PromptMAE achieves strong performance across all noise levels, maintaining high accuracy even at 75% noise where CE degrades drastically.

5.2. PromptOT: Optimal-Transport-Based Data Purification

Although PromptMAE provides robust training, CE loss retains advantages on clean data: it converges faster and achieves tighter class boundaries. To combine the strengths of both losses, we need a mechanism to distinguish clean from noisy samples. We propose PromptOT, which leverages optimal transport to globally assign pseudo-labels and partition the dataset.

Text-feature prototypes. A distinctive feature of PromptOT is its use of the text encoder's class-wise features as prototypes. Given the current learned prompts, we compute text features $\boldsymbol{T} = [\boldsymbol{t}_1, \ldots, \boldsymbol{t}_C] \in \mathbb{R}^{C \times d}$ by passing each class prompt through the frozen text encoder. These text features serve as semantically grounded class prototypes that are less susceptible to noise-induced drift compared to running-average-based prototypes.

Cost matrix construction. For each image $\boldsymbol{x}_i$ with normalized feature $\boldsymbol{z}_i$, we compute the similarity to all text pro-

totypes and construct the cost matrix:

$$\boldsymbol{M} = -\log(\boldsymbol{I} \cdot \boldsymbol{T}^\top), \tag{10}$$

where $\boldsymbol{I} \in \mathbb{R}^{N \times d}$ is the matrix of normalized image features and $\boldsymbol{T} \in \mathbb{R}^{C \times d}$ is the matrix of text features. The negative log-similarity serves as a transport cost: assigning an image to a class with high semantic similarity incurs low cost.

Entropic optimal transport. We formulate the pseudo-labeling problem as an entropic OT problem:

$$\begin{aligned} \min_{\boldsymbol{Q} \in \mathbb{R}^{N \times C}} \quad & \langle \boldsymbol{M}, \boldsymbol{Q} \rangle - \epsilon H(\boldsymbol{Q}), \\ \text{s.t.} \quad & \boldsymbol{Q}\mathbf{1}_C = \frac{1}{N}\mathbf{1}_N, \quad \boldsymbol{Q}^\top \mathbf{1}_N = \frac{\beta}{C}\mathbf{1}_C, \end{aligned} \tag{11}$$

where $\boldsymbol{Q}$ is the transport plan, $H(\boldsymbol{Q})$ is the entropic regularization, ϵ controls the regularization strength, and $\beta \in [0, 1]$ is a budget parameter controlling the fraction of samples assigned high-confidence labels. The marginal constraints ensure a balanced assignment across classes. We solve this efficiently using the Sinkhorn algorithm [10].

Curriculum scheduling. Following [5], we adopt a curriculum strategy where β increases over the training epochs from a small initial value to 1. In early epochs, only the most confident samples are selected, and as training progresses and the prompts improve, the budget expands to include more samples. The schedule follows:

$$\beta_t = \beta_0 + (1 - \beta_0) \cdot s\left(\frac{t}{T_c}\right), \tag{12}$$

where t is the current epoch, T_c is the curriculum duration, β_0 is the initial rate, and $s(\cdot)$ is a scheduling function (linear or exponential).

Data partitioning. Given the OT solution $\boldsymbol{Q}^*$, we derive pseudo-labels $\hat{y}_i = \arg\max_c \boldsymbol{Q}^*_{i,c}$. We identify high-confidence samples based on the OT coupling values and partition them as:

$$\mathcal{D}_{\text{clean}} = \{(\boldsymbol{x}_i, \tilde{y}_i) \mid \text{selected} \wedge \hat{y}_i = \tilde{y}_i\}, \tag{13}$$

$$\begin{aligned} \mathcal{D}_{\text{noisy}} = \{(\boldsymbol{x}_i, \tilde{y}_i) \mid \text{selected} \wedge \hat{y}_i \neq \tilde{y}_i\} \\ \cup \{(\boldsymbol{x}_i, \tilde{y}_i) \mid \text{not selected}\}. \end{aligned} \tag{14}$$

Samples whose OT-derived pseudo-labels agree with the given noisy labels are considered clean; those that disagree, or that were not selected by the budget constraint, are treated as noisy.

Algorithm 1 NLPrompt Training

Require: Training data $\mathcal{D} = \{(\boldsymbol{x}_i, \tilde{y}_i)\}_{i=1}^N$, pre-trained CLIP model, number of epochs T, curriculum parameters β_0, T_c

```
Initialize learnable context vectors {v_m}_{m=1}^M
for t = 1 to T do
    // PromptOT: Data Purification
    Compute text features T from current prompts
    Compute image features I for all training samples
    Construct cost matrix M = −log(I · T^⊤)
    Compute budget β_t via Eq. (12)
    Solve OT (Eq. (11)) to obtain Q*
    Partition into D_clean, D_noisy via Eqs. (13)-(14)
    // Harmonized Training
    for each mini-batch from D_clean do
        Update {v_m} with L_CE
    end for
    for each mini-batch from D_noisy do
        Update {v_m} with L_MAE
    end for
end for
```

5.3. NLPrompt: Harmonizing CE and MAE

The complete NLPrompt framework combines PromptMAE and PromptOT as follows. At the beginning of each epoch, PromptOT is executed to partition the current training data into $\mathcal{D}_{\text{clean}}$ and $\mathcal{D}_{\text{noisy}}$. Then, within the epoch, the prompt parameters are updated by:

$$\mathcal{L}_{\text{NLPrompt}} = \mathcal{L}_{\text{CE}}(\mathcal{D}_{\text{clean}}) + \mathcal{L}_{\text{MAE}}(\mathcal{D}_{\text{noisy}}), \tag{15}$$

where the two branches are applied sequentially: first, the clean subset is trained with CE to learn discriminative features efficiently, and then the noisy subset is trained with MAE to extract useful information without overfitting to incorrect labels. Both losses update the shared learnable prompt tokens via backpropagation through the frozen CLIP model.

The overall training procedure is summarized in Algorithm 1.

6. Experiments

6.1. Experimental Setup

Datasets. We evaluate NLPrompt on seven image classification benchmarks: Caltech101 [11], DTD [9], EuroSAT [17], Flowers102 [30], OxfordPets [31], StanfordCars [23], and UCF101 [34]. Following standard prompt learning protocols [45], we use 16-shot training data per class. For real-world noise evaluation, we use Food101N [24], which contains naturally noisy labels from web-crawled images.

Noise injection. We evaluate under both symmetric and asymmetric noise with rates $\eta \in \{12.5\%, 25.0\%, 37.5\%, 50.0\%, 62.5\%, 75.0\%\}$. Symmetric noise uniformly flips labels to any other class, while asymmetric noise flips labels only between semantically similar classes.

Baselines. We compare NLPrompt against: (1) CoOp [45], the standard prompt learning method with CE loss; (2) GCE, which replaces CE with the generalized cross-entropy loss [13]; and (3) JoAPR [15], a recent method that jointly optimizes prompts and refines labels for noisy prompt learning.

Implementation details. We use CLIP with a ViT-B/16 backbone as the pre-trained model. The number of context tokens is set to $M = 16$. We train for 20 epochs using SGD with a learning rate of 2×10^{-3}. The Sinkhorn regularization parameter is $\epsilon = 0.05$. The curriculum schedule uses a linear ramp from β_0 to 1 over the first half of training. All experiments are implemented in PyTorch [32] and repeated three times with different random seeds.

6.2. Main Results

Table 1 presents the comprehensive results across seven datasets under both symmetric and asymmetric noise at six noise levels. NLPrompt consistently achieves the best or near-best performance across nearly all settings.

On **Flowers102**, NLPrompt achieves 76.80% at 75% symmetric noise, outperforming GCE (70.37%) by 6.43 percentage points and JoAPR (66.93%) by 9.87 points. The gains are even more pronounced under asymmetric noise, where NLPrompt reaches 55.33% at 75% noise versus 39.23% for GCE and 13.27% for JoAPR. This demonstrates the effectiveness of combining MAE-based robust training with OT-based purification.

On **EuroSAT**, NLPrompt shows substantial improvements at high noise rates: 63.53% vs. 49.67% (GCE) at 62.5% symmetric noise and 32.73% vs. 17.50% (JoAPR) at 75% asymmetric noise. The satellite image classification task benefits particularly from PromptOT's global assignment, as the class boundaries are less intuitive for this domain.

On **StanfordCars**, while GCE is competitive at low noise rates (70.00% vs. 69.77% at 12.5% asymmetric), NLPrompt's advantage grows steadily with noise: at 75% symmetric noise, NLPrompt achieves 58.30% vs. 50.87% (GCE) and 39.40% (JoAPR). The fine-grained nature of car classification makes it particularly susceptible to noise, and NLPrompt's dual-loss strategy effectively mitigates this.

On **OxfordPets**, JoAPR achieves the best result at 75% symmetric noise (74.40% vs. NLPrompt's 70.77%), suggesting that for datasets with very strong visual-semantic alignment, JoAPR's label refinement can be highly effective at extreme noise. However, NLPrompt outperforms across all other noise levels and under asymmetric noise.

Across **UCF101** and **Caltech101**, NLPrompt consistently outperforms baselines, particularly at moderate to high noise rates. On DTD, the improvements are more modest but consistent, reflecting the challenge of texture recognition where even clean-label performance is lower.

6.3. Real-World Noisy Labels

Table 2 reports results on Food101N, a dataset with naturally occurring label noise from web images. NLPrompt achieves 76.46%, outperforming JoAPR (72.57%), GCE (71.32%), and CoOp (69.50%) by significant margins. This demonstrates that NLPrompt is effective not only under synthetic noise but also under the complex, instance-dependent noise patterns encountered in real-world datasets.

6.4. Ablation Studies

Table 3 presents ablation studies on Flowers102 under symmetric noise at rates of 10%, 30%, 50%, and 70%, isolating the contribution of each component.

Effect of loss function (rows a, b). Comparing CE (row a, 79.15% avg) with MAE (row b, 85.90% avg) on the full dataset without OT confirms the substantial advantage of MAE under noisy labels. MAE outperforms CE by 6.75 points on average, with the gap widening at higher noise rates (80.87% vs. 57.61% at 70% noise).

Effect of PromptOT (rows c–e vs. NLPrompt). Using OT without text-feature initialization (row c, 80.94% avg) degrades performance compared to even MAE alone, highlighting the importance of semantically grounded prototypes. Training only on clean data (row d, 82.16%) or only on noisy data (row e, 87.36%) each underperforms the full NLPrompt (92.00%), confirming the value of harmonizing both subsets.

Complementarity of components. The full NLPrompt achieves 92.00% on average, surpassing the best single-component variant (MAE alone, 85.90%) by 6.10 points. This demonstrates that PromptMAE and PromptOT are complementary: PromptOT identifies clean samples for efficient CE training, while PromptMAE extracts useful signal from noisy samples that would otherwise be discarded.

6.5. Generalization to Other Prompt Methods

Table 4 demonstrates that the NLPrompt framework generalizes beyond CoOp to other prompt learning methods. We apply our OT-based purification and dual-loss strategy

Table 1. Performance metrics across various datasets and noise levels. (%)

Dataset	Method	Noise Rate: Sym						Noise Rate: Asym					
		12.5%	**25.0%**	**37.5%**	**50.0%**	**62.5%**	**75.0%**	**12.5%**	**25.0%**	**37.5%**	**50.0%**	**62.5%**	**75.0%**
Flowers102	CoOp	88.93	83.50	77.93	70.10	55.60	37.17	86.97	74.70	60.43	42.60	26.53	12.60
	GCE	88.80	88.33	86.73	84.07	78.37	70.37	88.40	86.37	80.33	69.93	61.50	39.23
	JoAPR	85.57	81.23	74.60	70.23	67.90	66.93	85.17	79.63	73.97	73.83	53.37	13.27
	NLPrompt	**93.87**	**92.57**	**92.73**	**89.90**	**84.77**	**76.80**	**93.80**	**93.40**	**91.77**	**81.10**	**73.63**	**55.33**
DTD	CoOp	56.00	49.57	43.30	34.37	27.83	17.27	55.60	47.75	38.07	29.63	20.53	11.70
	GCE	61.00	59.83	56.80	50.73	43.60	33.67	60.70	57.57	52.70	43.97	33.40	18.23
	JoAPR	58.07	57.70	56.33	53.03	48.05	29.90	52.40	56.63	53.10	48.93	40.20	28.26
	NLPrompt	**62.97**	**61.23**	**59.17**	**55.17**	**49.03**	**39.80**	**62.30**	**60.60**	**56.47**	**50.80**	**40.27**	**28.37**
EuroSAT	CoOp	76.50	69.23	61.67	52.33	37.63	26.70	76.00	66.27	53.83	41.17	28.00	17.43
	GCE	82.13	78.60	74.67	63.13	49.67	31.40	78.23	72.70	63.63	45.30	22.90	12.10
	JoAPR	75.13	61.10	60.90	63.63	38.97	27.33	69.37	67.30	59.40	47.60	33.93	17.50
	NLPrompt	**82.53**	**79.53**	**78.13**	**66.70**	**63.53**	**43.80**	**80.13**	**77.13**	**71.43**	**54.30**	**37.27**	**32.73**
OxfordPets	CoOp	76.50	66.73	60.33	47.03	35.77	24.60	76.10	66.20	52.53	38.73	26.63	14.90
	GCE	85.63	84.60	83.67	79.23	71.40	53.17	85.50	83.03	76.73	68.07	50.70	31.97
	JoAPR	84.00	83.26	83.20	83.10	82.40	**74.40**	82.90	83.40	79.07	75.84	52.74	43.57
	NLPrompt	**86.17**	**86.00**	**85.33**	**84.87**	**83.63**	70.77	**86.00**	**84.97**	**82.40**	**77.53**	**66.33**	**48.60**
StanfordCars	CoOp	66.20	59.70	53.40	45.90	35.67	22.90	65.77	57.13	46.23	33.73	22.37	12.80
	GCE	**69.70**	66.40	66.47	63.77	59.25	50.87	**70.00**	66.45	61.23	53.67	39.65	26.60
	JoAPR	68.60	66.30	62.83	56.67	48.50	39.40	66.47	61.70	51.50	42.03	30.80	22.97
	NLPrompt	69.37	**68.80**	**67.20**	**65.63**	**62.83**	**58.30**	69.77	**67.53**	**64.23**	**59.03**	**50.90**	**39.50**
UCF101	CoOp	69.03	63.40	58.23	49.73	40.83	26.30	67.23	58.07	46.47	34.43	23.67	13.17
	GCE	74.00	**73.63**	72.57	69.37	66.00	57.07	73.90	71.87	67.97	62.23	52.50	36.37
	JoAPR	72.83	71.17	70.37	67.63	65.30	57.67	72.07	69.80	64.10	59.17	56.07	47.46
	NLPrompt	**74.83**	73.40	**72.83**	**70.33**	**68.10**	**60.53**	**74.90**	**73.53**	**71.03**	**65.97**	**58.97**	**49.27**
Caltech101	CoOp	86.43	81.03	76.73	70.90	61.33	46.90	84.93	75.23	62.87	49.43	33.57	20.33
	GCE	**92.00**	90.90	**90.80**	89.30	86.70	79.03	91.27	**91.20**	89.73	85.80	78.20	62.07
	JoAPR	90.30	90.45	89.90	88.27	86.93	83.93	90.30	89.30	88.30	88.73	85.80	81.90
	NLPrompt	91.73	**91.13**	90.77	**89.93**	**88.30**	**86.70**	**91.60**	91.17	**90.20**	**89.27**	**86.17**	**81.07**

Table 2. Test accuracy (%) on Food101N.

Method	CoOp	GCE	JoAPR	NLPrompt
Accuracy	69.50	71.32	72.57	76.46

Table 3. Ablation studies under multiple label noise ratios on Flowers102. (%)

	Method/Noise Ratio	10%	30%	50%	70%	Avg
w\o OT	(a) all data with CE	92.71	86.92	79.36	57.61	79.15
	(b) all data with MAE	88.47	89.07	85.20	80.87	85.90
w\ OT	(c) w\o text feature init	87.16	83.81	79.77	73.00	80.94
	(d) w\o noisy data	84.77	84.53	81.60	77.60	82.16
	(e) w\o clean data	90.17	90.13	88.60	80.55	87.36
	NLPrompt	**96.87**	**93.44**	**92.30**	**85.38**	**92.00**

to VPT [20], MaPLe [21], and PromptSRC [22] on Flowers102 under symmetric noise.

For VPT, adding NLPrompt improves accuracy from 27.67% to 73.90% at 75% noise, a gain of 46.23 points. MaPLe improves from 25.47% to 59.87% (+34.40 points), and PromptSRC from 49.37% to 59.50% (+10.13 points) at the same noise level. These results confirm that the noise-robustness gains of NLPrompt are orthogonal to the specific prompt architecture and can be readily integrated as a general-purpose module.

Table 4. The generalization of NLPrompt .

Method/Noise Ratio	12.5%	25.0%	37.5%	50.0%	62.5%	75.0%
VPT	89.20	79.43	65.20	61.37	41.67	27.67
VPT+Ours	**91.80**	**91.07**	**89.53**	**86.93**	**80.73**	**73.90**
MaPLe	83.27	77.67	65.27	55.40	37.53	25.47
MaPLe+Ours	**89.23**	**84.30**	**78.37**	**76.43**	**73.30**	**59.87**
PromptSRC	90.61	84.67	78.57	72.27	60.43	49.37
PromptSRC+Ours	**91.29**	**87.67**	**84.97**	**80.33**	**72.10**	**59.50**

6.6. Effect of Number of Shots

Figure 3 shows the effect of varying the number of shots per class on four datasets under 50% noise. NLPrompt consistently outperforms CoOp across all shot numbers and both noise types. The advantage is especially pronounced in low-shot settings (1–4 shots), where every mislabeled sample has an outsized impact. As the number of shots increases, both methods improve, but NLPrompt maintains a substantial lead, demonstrating its effectiveness across the few-shot

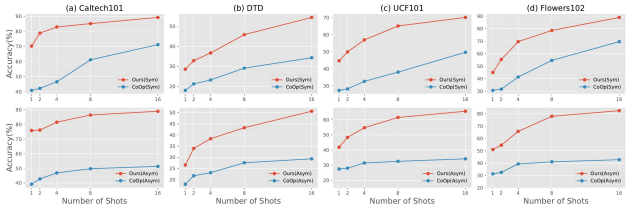

Figure 3. Performance comparison between NLPrompt (Ours) and CoOp under symmetric (top row) and asymmetric (bottom row) noise at 50% noise rate with varying numbers of shots on four datasets. NLPrompt consistently outperforms CoOp across all shot settings, with the gap being particularly large in low-shot regimes.

spectrum.

7. Conclusion

We presented NLPrompt, a unified framework for noise-robust prompt learning in vision-language models. NLPrompt integrates two complementary components: PromptMAE, which leverages the inherent noise tolerance of MAE loss for prompt optimization, and PromptOT, which uses entropic optimal transport with text-feature prototypes to partition training data into clean and noisy subsets. Through theoretical analysis grounded in feature learning theory, we showed that MAE improves the signal-to-noise ratio during prompt training by suppressing the gradient contribution of mislabeled samples. The harmonized dual-loss training strategy—CE on clean data and MAE on noisy data—combines the strengths of both losses. Extensive experiments across seven benchmarks under synthetic noise and one real-world noisy dataset demonstrate that NLPrompt consistently outperforms existing methods, often by large margins. The framework further generalizes to diverse prompt learning architectures including VPT, MaPLe, and PromptSRC, establishing NLPrompt as a broadly applicable solution for robust prompt learning under label noise.

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